

Carlini and Wagner Paper | (CW Attack) – Nicholas Wagner and David Wagner (2017)

CW attack is a type of **white-box** adversarial attacks that propose **Targeted** attacks based on 3 different norm metrics: $-L_\infty$ attack $-L_2$ attack $-L_0$ attack

At time of publishment of paper, they were strongest attacks and they have 3 advantages including: Minimum amount of perturbation – Resistance to defenses – Transferable across different DL models.

Given input X , the assigned class in NN is: $C(x) = \operatorname{argmax}_i(y_i) = \operatorname{argmax}_i(F(x)_i)$ where true label is denoted by $C(x)^*$, and the last layer of NN performs a softmax then we have logits: $F(x) = \operatorname{Softmax}(Z(y_i))$. They are trying to create a input X' that is similar to X , such that $C(X')=t$ where t is not true label.

Carlini and Wagner propose following formulation as optimization equation for attack:

$$\text{Minimize } D(x, x + \delta) \text{ such that } f(x + \delta) \leq 0 \text{ and } X + \delta \in [0, 1]^n$$

And they use lagrange multiplies to reform the problem into:

$$\text{Minimize } D(x, x + \delta) + C \cdot f(x + \delta)$$

And they chose the 6th variant of f : (known as margin loss function):

$$f(x') = \max(\max\{Z(x')_i \mid i \neq t\} - Z(x')_t, -k)$$

Where $\max\{Z(x')_i\}$ refers to maximum logit values of other class i than the target class t . The function calculates the difference in the logits between the target class t and the closest-to-the-target class. And k is confidence value that authors consider it as 0.

$-L_\infty$ attack: In this norm the distance is: $D(x, x + \delta) = \|\delta\|_\infty$ and it means the pixel in x' with the largest change from x : $\text{Minimize } \|\delta\|_\infty + c \cdot f(x + \delta)$ but as $\|\delta\|_\infty$ has poor optimization results. So: $\text{Minimize } \sum_i [(\delta_i - \tau)^+] + c \cdot f(x + \delta)$ and τ is set to 1 and decreases by rounds.

$-L_2$ attack: In this norm the distance is: $D(x, x + \delta) = \|\delta\|_2$ and we use parameter w as box-constraint: $\text{minimize } \|\delta\|_2^2 + c \cdot f(x + \delta)$ where $\delta = \frac{1}{2}(\tanh(w) + 1) - x$

$$\text{minimize } \left\| \frac{1}{2}(\tanh(w) + 1) - x \right\|_2^2 + c \cdot f\left(\frac{1}{2}(\tanh(w) + 1)\right)$$

$-L_0$ attack: Number of non-zero pixels in δ . They propose a iterative approach where at each iteration goal is to find the pixels that or not important and don't have much effect on classifier's output: Initialization: All allowed set of pixels in the image. Perform L_2 attack to find adv example $x + \delta$. Compute the gradient $g = \nabla f(x + \delta)$, where f is objective function in L_2 . Identify last important pixel $i = \operatorname{argmin}_i g_i$ and remove that pixel from allowed set. Iterate until L_2 attack fails to find adv example.